S181M116 Derivatives and Alternative Investments: Lecture 4

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This lecture introduces options as asymmetric derivative contracts and develops the first no-arbitrage restrictions on their prices. We begin with option market mechanics, option positions, contract specifications, exchange trading, margin, clearing, and over-the-counter markets. We then study the determinants of option prices, upper and lower bounds, put-call parity, and early exercise. The final part introduces binomial trees, risk-neutral valuation, American exercise, delta, and the link between tree parameters and volatility. Sections 13.10 and 13.11 from the source material are intentionally omitted.

Reading. Hull, Chapters 10, 11, and Chapter 13 through Section 13.9. The practical goal is to read option contracts and option quotes correctly. The pricing goal is to see how no-arbitrage turns payoffs into restrictions, relations, and eventually full valuation methods.

Roadmap

Lectures 1–3 studied forwards, futures, and interest-rate instruments. Those contracts are symmetric commitments: both sides are obligated to transact later. Options introduce a different structure. The holder has a right but not an obligation, while the writer has a contingent obligation.

The lecture has three blocks:

1. **Option contracts and markets.** Define calls, puts, long and short positions, moneyness, contract specifications, margin, clearing, and trading arrangements.
2. **Static no-arbitrage restrictions.** Derive bounds and parity relations before introducing a full probability model.
3. **Binomial valuation.** Price options by building riskless portfolios and by using risk-neutral probabilities on one-step and multi-step trees.

Main lesson. The option premium is not the expected payoff discounted at an arbitrary risky discount rate. No-arbitrage prices options by comparing them with dynamically chosen positions in the underlying asset and the risk-free asset.

Option Basics

Rights versus Obligations

An option gives its holder a choice. A forward or futures contract locks both parties into an exchange, but an option allows the holder to walk away when exercise is unfavorable.

Definition 1 (Call option)
A call option gives the holder the right to buy the underlying asset at a fixed strike price K on or before a fixed maturity date.

Definition 2 (Put option)
A put option gives the holder the right to sell the underlying asset at a fixed strike price K on or before a fixed maturity date.

Definition 3 (Strike and maturity)
The strike price*, or exercise price, is the price at which the underlying asset can be bought or sold if the option is exercised. The* maturity *or expiration date is the last date relevant for exercise.*

The right is valuable because exercise is selective. A call holder exercises only when buying at K is better than buying in the market. A put holder exercises only when selling at K is better than selling in the market. The writer receives the premium up front, but must honor the holder's choice.

European and American Exercise

Definition 4 (European option)
A European option can be exercised only at maturity.

Definition 5 (American option)
An American option can be exercised at any time up to and including maturity.

These names describe exercise style, not geography. European options are easier to analyze because the exercise date is fixed. American options are usually more valuable, or at least no less valuable, because they include all European exercise opportunities plus the possibility of early exercise.

$$C \geq c, \qquad P \geq p,$$

where C and P denote American call and put prices, and c and p denote European call and put prices with the same underlying, strike, and maturity.

Payoff versus Profit

The payoff is the value received at expiration before subtracting the initial premium. The profit subtracts the initial premium, usually ignoring the time value of money in payoff diagrams.

For a European call:

$$\text{payoff} = \max(S_T - K, 0).$$

For a European put:

$$\text{payoff} = \max(K - S_T, 0).$$

If the call premium is c_0, the call profit at maturity is

$$\max(S_T - K, 0) - c_0.$$

If the put premium is p_0, the put profit at maturity is

$$\max(K - S_T, 0) - p_0.$$

> **Exercise can reduce a loss.** Suppose a call has strike 100 and was bought for 5. If the stock finishes at 102, the holder exercises. The payoff is 2 and the total profit is -3. Exercising does not make the trade profitable, but it is better than letting the option expire worthless and losing 5.

The Four Basic Option Positions

Every option has a buyer and a writer. The buyer is long the option; the writer is short the option.

Long Call

A long call benefits from increases in the underlying price. Its payoff is

$$\max(S_T - K, 0).$$

The downside is limited to the premium. The upside is potentially unlimited for a stock because the stock price can in principle rise without bound.

Short Call

A short call is the writer's side of the same contract:

$$-\max(S_T - K, 0) = \min(K - S_T, 0).$$

The writer receives the premium, but faces losses when the underlying price rises above the strike. A naked short call is therefore a high-risk position.

Long Put

A long put benefits from decreases in the underlying price:

$$\max(K - S_T, 0).$$

The downside is limited to the premium. The payoff is largest when the underlying price is very low. For a stock, the maximum payoff is approximately K because the stock price cannot fall below zero.

Short Put

A short put has payoff

$$-\max(K - S_T, 0) = \min(S_T - K, 0).$$

The writer receives the premium and loses when the underlying price falls below the strike. The writer may be forced to buy the asset at K even when its market value is much lower.

> **Mnemonic.** Long option positions have rights and pay premiums. Short option positions receive premiums and carry contingent obligations.

Underlying Assets and Contract Design

Stock Options

Exchange-traded stock options are standardized contracts. In the United States, one stock option contract typically covers 100 shares. A quoted option price is usually the price per share, so a quote of 4.20 corresponds to a premium of

$$4.20 \times 100 = 420$$

for one contract.

Other Underlyings

Options can be written on many assets:

Exchange-traded products. Options on ETFs and other listed products allow investors to take option exposure to broad equity, bond, commodity, or currency baskets.

Foreign currencies. Currency options give the right to exchange one currency for another at a specified exchange rate.

Stock indices. Index options settle in cash because delivering the entire index portfolio would be inconvenient.

Futures contracts. A futures option gives the right to enter a futures position at the strike. Exercise gains depend on the futures price relative to the strike.

Expiration Dates and Strike Prices

An exchange must specify which option contracts trade. For stock options this includes the underlying stock, exercise style, contract size, expiration date, strike price, and adjustment rules.

Standard U.S. equity options often expire on the third Friday of the expiration month, while weekly options expire on other Fridays. Longer-dated equity options are often called LEAPS. Strike spacing is chosen by exchanges and is typically finer for low-price stocks and wider for high-price stocks.

Option Class and Option Series

Definition 6 (Option class)
An option class consists of all options of the same type on the same underlying, such as all IBM calls or all IBM puts.

Definition 7 (Option series)
An option series consists of options of a given class with the same strike and same expiration date, such as October calls with strike 160.

This distinction matters operationally. A trader may discuss the whole option class when talking about liquidity in a stock's options, but a trade is in a specific option series.

Moneyness

Moneyness describes whether immediate exercise would produce a positive payoff. For stock price S and strike K:

	Call	Put
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$

Definition 8 (Intrinsic value)
The intrinsic value is the payoff from immediate exercise:

$$\textit{call intrinsic value} = \max(S - K, 0),$$
$$\textit{put intrinsic value} = \max(K - S, 0).$$

Definition 9 (Time value)
The time value of an option is its market price minus its intrinsic value.

An American option cannot sell for less than its intrinsic value because the holder can exercise immediately. A European option need not satisfy this immediate-exercise bound because immediate exercise is not allowed.

Dividends, Stock Splits, and Contract Adjustments

Exchange-traded stock options are usually not adjusted for ordinary cash dividends. The stock price falls on the ex-dividend date, and option prices reflect expected dividends through valuation.

By contrast, stock splits and stock dividends generally lead to contract adjustments. The goal is to keep the economic position of the option buyer and writer unchanged.

Example 1 (Two-for-one stock split)
A call option gives the holder the right to buy 100 shares at 30. After a two-for-one stock split, the option is adjusted to cover 200 shares at a strike of 15. The total exercise cost remains

$$100 \times 30 = 200 \times 15 = 3000.$$

Example 2 (Stock dividend)
A put option gives the holder the right to sell 100 shares at 15. A 25% stock dividend is equivalent to a 5-for-4 split. The adjusted option covers 125 shares at a strike of 12, keeping total exercise proceeds equal to

$$100 \times 15 = 125 \times 12 = 1500.$$

Trading, Margin, Clearing, and Regulation

Market Makers and Bid-Ask Spreads

Option markets require liquidity across many strikes and maturities. Market makers quote bid and ask prices and stand ready to trade at those quotes, within specified size limits. They earn compensation through the bid-ask spread and manage inventory risk by hedging.

$$\text{bid} \leq \text{mid} \leq \text{ask}.$$

For illiquid option series, the spread can be large relative to the option's premium. This is especially important for low-priced out-of-the-money options.

Offsetting Trades

Most traders close option positions by entering an offsetting order rather than by exercising. A trader who bought a call can sell the same call series. A trader who wrote a put can buy back the same put series. The economic exposure is then closed through the market.

Trading Costs

Option trades involve more than the model price:

- bid-ask spreads;
- brokerage commissions and exchange fees;
- market impact for large orders;
- financing and collateral costs;
- tax and accounting treatment, where relevant.

Trading costs are small in many textbook derivations, but they matter for strategy design and for deciding whether an apparent arbitrage can actually be implemented.

Margin for Option Writers

Buying an option creates no future obligation beyond the premium already paid. Writing an option creates a possible future obligation, so brokers and clearing organizations require margin from writers.

For a naked option, the margin protects against adverse movements in the underlying. For covered or spread positions, margin can be lower because another position partly offsets the written option's exposure.

> **Economic interpretation.** Margin is not the option's price. It is collateral against future performance. The writer keeps the option premium but must post enough collateral to reduce counterparty and clearing risk.

The Options Clearing Corporation

For exchange-traded options in the United States, the Options Clearing Corporation acts as the central clearing counterparty. It records positions, handles exercises, and guarantees performance by standing between buyers and writers after trades are cleared.

This clearing structure makes exchange-traded options different from bilateral over-the-counter options. In an OTC contract, the option holder faces the credit risk of the writer unless the contract is collateralized or centrally cleared.

Exercise Assignment

When an option holder exercises, the clearing system assigns the exercise to a writer. The writer usually does not know in advance whether a particular short option will be assigned, only that assignment can occur if the option is exercised.

Position and Exercise Limits

Exchanges impose position limits and exercise limits to reduce the chance that a single trader can dominate a market or create disorderly exercise activity. Long calls and short puts are typically treated as one side of the market because both benefit from price increases. Short calls and long puts are treated as the opposite side.

Taxation and Related Instruments

Tax rules can materially affect option trading incentives. Wash sale rules, constructive sale rules, and the classification of option gains can matter for real investors. These rules are jurisdiction-specific and can change, so they should be treated as institutional background rather than as universal pricing logic.

Options are also related to warrants, employee stock options, and convertible bonds:

Warrants are call-like securities issued by a company. Exercise usually creates new shares and can dilute existing shareholders.

Employee stock options are compensation contracts designed to align incentives. They are often at-the-money at grant and subject to vesting restrictions.

Convertible bonds combine debt with an embedded option to convert into equity.

Over-the-Counter Options

OTC options are negotiated bilaterally and can be tailored to a client's exact needs. They may differ from exchange-traded options in strike, maturity, underlying, exercise dates, payoff shape, collateral terms, and settlement rules. Tailoring is valuable, but it introduces greater counterparty credit, documentation, model, and liquidity risk.

Factors Affecting Option Prices

The Six Main Inputs

The value of a stock option depends on:

1. current stock price S_0;
2. strike price K;
3. time to expiration T;
4. stock price volatility σ;
5. risk-free interest rate r;
6. dividends expected during the option's life.

These inputs affect calls and puts differently.

Increase in input	Call value	Put value
Current stock price S_0	Increases	Decreases
Strike price K	Decreases	Increases
Volatility σ	Increases	Increases
Risk-free rate r	Usually increases	Usually decreases
Expected dividends	Decreases	Increases

Stock Price and Strike Price

A call is more valuable when the stock price is higher and less valuable when the strike is higher:

$$\max(S_T - K, 0)$$

increases with S_T and decreases with K.

A put has the opposite directional exposure:

$$\max(K - S_T, 0)$$

increases with K and decreases with S_T.

Time to Expiration

For American options, more time cannot reduce value because the longer-dated option contains every exercise opportunity available to the shorter-dated option. Thus, for otherwise identical American options, more time generally means a weakly higher price.

For European options, the effect of time is subtler. A longer-maturity European option does not permit exercise at the shorter maturity. Large expected dividends or interest-rate effects can make a shorter European option more valuable in special cases.

Volatility

Volatility increases both call and put values. An option has limited downside for the holder but retains upside from favorable movements. Higher volatility therefore increases the chance that the option finishes deep in the money.

This does not mean volatility makes the underlying asset itself more valuable. For a stock holder, upside and downside movements both matter. For an option holder, the payoff truncates the bad side.

Risk-Free Interest Rate

Holding the stock price fixed, a higher risk-free rate tends to increase call values and decrease put values. Intuitively:

- a call delays payment of the strike, so a higher rate makes that delayed payment less costly in present-value terms;
- a put delivers the strike in the future, so a higher rate reduces the present value of that future receipt.

Dividends

Expected dividends reduce the stock price when the stock goes ex-dividend. This hurts calls and benefits puts, all else equal. Dividend timing is also central to early exercise decisions for American calls.

Assumptions and Notation for No-Arbitrage Results

We use the following notation:

$$
\begin{aligned}
S_0 &= \text{current stock price,}\\
K &= \text{strike price,}\\
T &= \text{time to maturity,}\\
r &= \text{continuously compounded risk-free rate,}\\
c &= \text{European call price,}\\
p &= \text{European put price,}\\
C &= \text{American call price,}\\
P &= \text{American put price.}
\end{aligned}
$$

The standard static no-arbitrage arguments assume that large market participants can borrow and lend at the risk-free rate, short sell where needed, trade without material transaction costs, and exploit arbitrage opportunities quickly.

Role of assumptions. These assumptions are not claims that markets are frictionless in every detail. They identify benchmark restrictions. If a price violates them by more than trading costs and constraints, arbitrage pressure should push it back.

Upper and Lower Bounds

Upper Bounds

A call option cannot be worth more than the underlying stock:

$$c \leq S_0, \qquad C \leq S_0.$$

The call is a right to buy the stock, not ownership of more than the stock.

A European or American put cannot be worth more than the strike:

$$p \leq K, \qquad P \leq K.$$

For a European put, a sharper bound is

$$p \leq Ke^{-rT},$$

because the maximum payoff K occurs at maturity and must be discounted.

Lower Bound for a European Call

For a non-dividend-paying stock, a European call satisfies

$$c \geq S_0 - Ke^{-rT}.$$

It also cannot have negative value, so

$$c \geq \max(S_0 - Ke^{-rT}, 0).$$

Proof 1
Compare two portfolios:

Portfolio A: one European call plus cash Ke^{-rT};

Portfolio B: one share of stock.

At maturity, Portfolio A is worth

$$\max(S_T - K, 0) + K.$$

If $S_T > K$, this is S_T. If $S_T \leq K$, this is K, which is at least S_T. Portfolio A is therefore always worth at least as much as Portfolio B at maturity. It must be worth at least as much today:

$$c + Ke^{-rT} \geq S_0.$$

Example 3 (Call lower bound)
Suppose $S_0 = 20$, $K = 18$, $r = 10\%$, and $T = 1$. Then

$$c \geq 20 - 18e^{-0.10} \approx 20 - 16.29 = 3.71.$$

If the call traded for less than 3.71, buying the call and lending the present value of the strike would dominate owning the stock.

Lower Bound for a European Put

For a non-dividend-paying stock, a European put satisfies

$$p \geq Ke^{-rT} - S_0.$$

Combining with non-negativity:

$$p \geq \max(Ke^{-rT} - S_0, 0).$$

Proof 2
Compare:

Portfolio A: one European put plus one share;

Portfolio B: cash Ke^{-rT}.

At maturity, Portfolio A is

$$\max(K - S_T, 0) + S_T.$$

If $S_T < K$, this equals K. If $S_T \geq K$, this equals S_T, which is at least K. Portfolio A dominates a risk-free payoff of K, so

$$p + S_0 \geq Ke^{-rT}.$$

Example 4 (Put lower bound)
Suppose $S_0 = 37$, $K = 40$, $r = 5\%$, and $T = 0.5$. Then

$$p \geq 40e^{-0.05(0.5)} - 37 \approx 39.01 - 37 = 2.01.$$

Put-Call Parity

European Parity for Non-Dividend-Paying Stocks

Put-call parity is the central static-arbitrage relation for European options:

$$c + Ke^{-rT} = p + S_0.$$

The left side is a fiduciary call: a call plus enough cash to pay the strike at maturity. The right side is a protective put: a stock plus a put. Both portfolios pay exactly

$$\max(S_T, K)$$

at maturity.

State at T	Call + cash	Stock + put	Common payoff
$S_T > K$	$S_T - K + K$	$S_T + 0$	S_T
$S_T \leq K$	$0 + K$	$S_T + K - S_T$	K

Because the terminal payoffs are identical, the current values must be identical in an arbitrage-free market.

Using Parity to Detect Mispricing

Parity can be rewritten as

$$c - p = S_0 - Ke^{-rT}.$$

The difference between the call and put price equals the forward-like value of buying the stock and financing the strike.

Example 5 (Parity-implied put price)
Suppose $S_0 = 31$, $K = 30$, $r = 6\%$, $T = 0.25$, and the European call costs 3. Then parity gives

$$p = c + Ke^{-rT} - S_0 = 3 + 30e^{-0.06(0.25)} - 31 \approx 1.55.$$

Arbitrage Logic When Parity Fails

If

$$c + Ke^{-rT} > p + S_0,$$

the fiduciary call is too expensive relative to the protective put. An arbitrage trader sells the expensive side and buys the cheap side:

$$\text{short call, borrow } Ke^{-rT}, \text{ buy put, buy stock.}$$

If

$$c + Ke^{-rT} < p + S_0,$$

the protective put is too expensive relative to the fiduciary call. The trader buys the fiduciary call and sells the protective put:

$$\text{buy call, lend } Ke^{-rT}, \text{ short put, short stock.}$$

American Options and Parity Inequalities

American options allow early exercise, so exact European parity does not generally hold. For a non-dividend-paying stock, the following inequality is the standard restriction:

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}.$$

The upper bound is related to the European parity relation and the fact that an American put is at least as valuable as a European put. The lower bound reflects the immediate-exercise value embedded in American contracts.

Early Exercise

American Calls on Non-Dividend-Paying Stocks

It is never optimal to exercise an American call early when the stock pays no dividends. The holder gives up two valuable things by exercising:

1. the time value of the option;
2. the interest that could be earned by delaying payment of the strike.

The call holder can participate in upside without paying the strike until exercise. Exercising early converts the option into stock and unnecessarily pays K before it is required.

$$C = c$$

for a non-dividend-paying stock when the call is otherwise identical.

American Puts on Non-Dividend-Paying Stocks

Early exercise of an American put can be optimal. A deep-in-the-money put pays approximately $K - S$ when exercised. If S is very low, exercising allows the holder to receive K early and earn interest on it.

This early-exercise possibility makes American puts more valuable than European puts in some cases:

$$P \geq p.$$

The American put lower bound is its intrinsic value:

$$P \geq \max(K - S_0, 0).$$

Effect of Dividends on Early Exercise

Dividends can make early exercise of American calls optimal. A call holder does not receive dividends unless the option is exercised and converted into stock. Just before a large ex-dividend date, a deep-in-the-money American call may be exercised to capture the dividend.

For puts, dividends make the underlying stock less valuable after the ex-dividend date, which can reduce the incentive to exercise immediately before the dividend. The exact early-exercise decision depends on interest rates, dividends, moneyness, and remaining time value.

Dividends and Static Bounds

Known Cash Dividends

Let D be the present value of known cash dividends paid during the option's life. A stock plus its dividends is economically different from a stock without dividends. The no-arbitrage bounds adjust by subtracting D from the stock component.

For European calls:

$$c \geq \max(S_0 - D - Ke^{-rT}, 0).$$

For European puts:

$$p \geq \max(Ke^{-rT} + D - S_0, 0).$$

Put-Call Parity with Known Dividends

With known dividends, European put-call parity becomes

$$c + D + Ke^{-rT} = p + S_0,$$

or equivalently

$$c - p = S_0 - D - Ke^{-rT}.$$

Interpretation. The dividend-adjusted stock price $S_0 - D$ is the part of the stock value that remains attached to the share price at option maturity. The dividends paid before maturity are not received by the option holder unless the option is exercised.

One-Step Binomial Model

Tree Setup

The binomial model assumes that over a short period the stock price has only two possible movements. Starting from S_0, the stock moves to

$$S_u = S_0 u \qquad \text{or} \qquad S_d = S_0 d,$$

where

$$u > 1, \qquad d < 1.$$

Let the option value after an up move be f_u and after a down move be f_d. The goal is to compute the current option value f.

Riskless Portfolio Argument

Construct a portfolio with:

$$\Delta \text{ shares of stock} \quad \text{and} \quad -1 \text{ option.}$$

Its value after an up move is

$$S_0 u\Delta - f_u.$$

Its value after a down move is

$$S_0 d\Delta - f_d.$$

Choose Δ so the two values are equal:

$$S_0 u\Delta - f_u = S_0 d\Delta - f_d.$$

Solving gives

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}.$$

Definition 10 (Delta in a one-step tree)
Delta is the number of shares needed per option shorted to make the stock-option portfolio locally riskless:

$$\Delta = \frac{\textit{change in option value}}{\textit{change in stock value}}.$$

Because the portfolio is riskless, it must earn the risk-free rate. Discounting the certain terminal value back to today determines f.

Risk-Neutral Probability

The same value can be written in risk-neutral form:

$$f = e^{-rT}\left[pf_u + (1-p)f_d\right],$$

where

$$p = \frac{e^{rT} - d}{u - d}.$$

The parameter p is not generally the real-world probability of an up move. It is the probability that makes the expected stock return equal to the risk-free rate:

$$pS_0 u + (1-p)S_0 d = S_0 e^{rT}.$$

> **Risk-neutral valuation.** To price the option, replace the stock's expected return by the risk-free rate, compute the expected option payoff under that adjusted probability, and discount at the

risk-free rate.

Numerical One-Step Example

Suppose

$$S_0 = 20, \quad u = 1.10, \quad d = 0.90, \quad K = 21, \quad r = 12\%, \quad T = 0.25.$$

The possible stock prices are

$$S_u = 22, \qquad S_d = 18.$$

For a European call,

$$f_u = \max(22 - 21, 0) = 1, \qquad f_d = \max(18 - 21, 0) = 0.$$

The risk-neutral probability is

$$p = \frac{e^{0.12(0.25)} - 0.90}{1.10 - 0.90} \approx \frac{1.03045 - 0.90}{0.20} = 0.6523.$$

The option value is

$$f = e^{-0.12(0.25)}[0.6523(1) + 0.3477(0)] \approx 0.633.$$

The hedge ratio is

$$\Delta = \frac{1 - 0}{22 - 18} = 0.25.$$

Thus a trader who is short one call hedges the one-step risk by buying 0.25 shares.

Why the Real-World Expected Return Drops Out

The option is risky, so it is tempting to discount its expected payoff using a risky expected return. The binomial argument shows why this is not the right primitive. The option is priced relative to the stock. Once the stock price S_0 is observed, the stock already embeds the market's required compensation for risk.

The no-arbitrage relation between the option and the stock does not require the stock's expected real-world return μ. Different investors may disagree about μ, but if they agree on S_0, u, d, r, and the payoff, the arbitrage-free option price is pinned down.

Key distinction. Risk-neutral valuation does not say investors are risk-neutral in reality. It says that for pricing derivatives relative to traded underlyings, we may use risk-neutral proba-

bilities because the risk adjustment is already reflected in traded asset prices.

Two-Step Binomial Trees

Backward Induction

In a two-step tree, the stock can move up or down in each step. Starting from S_0, the terminal stock prices are

$$S_0u^2, \qquad S_0ud, \qquad S_0d^2.$$

If $ud = du$, the middle node is recombining: one up then one down gives the same stock price as one down then one up.

The valuation procedure is:

1. compute terminal option payoffs;
2. value the option one step earlier using the one-step formula;
3. continue backward until reaching the initial node.

For time step Δt, the one-step recursion is

$$f = e^{-r\Delta t}\left[pf_u + (1-p)f_d\right],$$

where

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Two-Step Formula for a European Option

Let the terminal option payoffs be f_{uu}, f_{ud}, and f_{dd}. Then

$$\begin{aligned} f_u &= e^{-r\Delta t}\left[pf_{uu} + (1-p)f_{ud}\right], \\ f_d &= e^{-r\Delta t}\left[pf_{ud} + (1-p)f_{dd}\right], \\ f &= e^{-r\Delta t}\left[pf_u + (1-p)f_d\right]. \end{aligned}$$

Substituting gives

$$f = e^{-2r\Delta t}\left[p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}\right].$$

European Put Example

Consider a two-year European put with

$$S_0 = 50, \quad K = 52, \quad u = 1.2, \quad d = 0.8, \quad r = 5\%, \quad \Delta t = 1.$$

The risk-neutral probability is

$$p = \frac{e^{0.05} - 0.8}{1.2 - 0.8} \approx 0.6282.$$

The terminal stock prices are

$$72, \quad 48, \quad 32.$$

The terminal put payoffs are

$$f_{uu} = 0, \quad f_{ud} = 4, \quad f_{dd} = 20.$$

Thus

$$\begin{aligned} f &= e^{-0.10}\left[(0.6282)^2(0) + 2(0.6282)(0.3718)(4) + (0.3718)^2(20)\right] \\ &\approx 4.19. \end{aligned}$$

American Options in Trees

Exercise Check at Each Node

For American options, backward induction must include an exercise decision at every node before maturity. At a node with stock price S, the value is

$$\max\{\text{intrinsic value}, \text{continuation value}\}.$$

For an American put:

$$\text{node value} = \max\left[K - S,\, e^{-r\Delta t}\left(pf_u + (1-p)f_d\right)\right].$$

For an American call:

$$\text{node value} = \max\left[S - K,\, e^{-r\Delta t}\left(pf_u + (1-p)f_d\right)\right].$$

Early Exercise Example Logic

Return to the put with $S_0 = 50$, $K = 52$, $u = 1.2$, $d = 0.8$, $r = 5\%$, and $\Delta t = 1$. At the lower node after one down move, the stock price is 40. Immediate exercise gives

$$K - S = 52 - 40 = 12.$$

The continuation value is the discounted risk-neutral expectation of next period's values. If continuation is less than 12, the American put is exercised at that node.

Backward induction with American exercise. First compute continuation value as if the option were European from that node onward. Then compare it with immediate exercise. The node value is the larger of the two.

Delta Hedging

Delta as Local Sensitivity

Delta measures the sensitivity of the option price to the underlying price:

$$\Delta = \frac{\partial f}{\partial S}$$

in continuous models, and

$$\Delta = \frac{f_u - f_d}{S_u - S_d}$$

in a one-step binomial tree.

Call deltas are positive. Put deltas are negative. A deeply in-the-money call has delta close to 1, while a deeply out-of-the-money call has delta close to 0. A deeply in-the-money put has delta close to -1, while a deeply out-of-the-money put has delta close to 0.

Delta Changes Over Time

In a multi-step tree, delta is not constant. It differs by node because the option's moneyness and remaining time differ by node. A hedge that is riskless over one small interval must be rebalanced as the stock price and time to maturity change.

Dynamic hedging. Option hedging is dynamic. The portfolio that is riskless over one short interval is not usually riskless over the next interval unless the stock position is adjusted.

Matching Binomial Trees to Volatility

Choosing u and d

A binomial tree needs three parameters per time step:

$$u, \quad d, \quad p.$$

Once u and d are chosen, no-arbitrage fixes p:

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

The up and down multipliers are chosen to match volatility. If annual volatility is σ and the time step is Δt, the standard binomial choice is

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}.$$

Then

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Why Volatility Is Preserved Under Risk-Neutral Valuation

Moving from the real-world measure to the risk-neutral measure changes expected returns. The real-world expected stock growth rate μ is replaced by the risk-free rate r. Volatility is not changed. The tree should therefore match the asset's volatility under both the real-world and risk-neutral measures, but use a risk-neutral probability when pricing.

> **Connection to measure changes.** In more advanced notation, the real-world probability measure is often called P, while the risk-neutral measure is called Q. The change from P to Q changes drifts, not the local volatility parameter used to build the tree.

Numerical Tree Parameters

Suppose a two-year option is valued with two time steps, so $\Delta t = 1$. Let

$$\sigma = 30\%, \qquad r = 5\%.$$

Then

$$u = e^{0.30} = 1.3499, \qquad d = e^{-0.30} = 0.7408,$$

and

$$p = \frac{e^{0.05} - 0.7408}{1.3499 - 0.7408} \approx 0.5097.$$

These parameters define the tree. For an American option, each node then uses the continuation-versus-exercise comparison.

Increasing the Number of Steps

More Steps Means a Finer Approximation

The same tree formulas apply for any number of steps n. If the option life is T, then

$$\Delta t = \frac{T}{n}.$$

The parameters are

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}}, \qquad p = \frac{e^{r\Delta t} - d}{u - d}.$$

As n increases, the tree has a finer time grid and a richer set of possible terminal stock prices. For European options on non-dividend-paying stocks, the binomial price converges to the Black-Scholes-Merton price as the number of steps becomes large.

Five-Step Illustration

Suppose $T = 2$, $n = 5$, $r = 5\%$, and $\sigma = 30\%$. Then

$$\Delta t = \frac{2}{5} = 0.4.$$

The tree parameters are

$$u = e^{0.30\sqrt{0.4}} \approx 1.2089, \qquad d = \frac{1}{u} \approx 0.8272,$$

$$e^{r\Delta t} = e^{0.05(0.4)} \approx 1.0202, \qquad p = \frac{1.0202 - 0.8272}{1.2089 - 0.8272} \approx 0.5056.$$

The pricing algorithm does not change. Compute terminal payoffs, work backward one step at a time, and for American options compare continuation value with exercise value at each node.

What Is Omitted

The source material includes Section 13.10 on DerivaGem software and Section 13.11 on options on other assets. These notes intentionally exclude those two sections. The binomial-tree discussion here is limited to stock options through the increasing-number-of-steps material in Section 13.9.

Worked Examples

Example 6 (Payoff and profit of a long call)
A call has strike 100 and premium 6. At maturity the stock price is 114. The payoff is

$$\max(114 - 100, 0) = 14.$$

The profit per share is

$$14 - 6 = 8.$$

For one standard contract on 100 shares, the profit is 800.

Example 7 (Payoff and profit of a long put)
A put has strike 70 and premium 4. At maturity the stock price is 58. The payoff is

$$\max(70 - 58, 0) = 12.$$

The profit per share is

$$12 - 4 = 8.$$

Example 8 (Intrinsic and time value)
A call has strike 50, the stock price is 56, and the call price is 8.50. The intrinsic value is

$$\max(56 - 50, 0) = 6.$$

The time value is

$$8.50 - 6 = 2.50.$$

Example 9 (European parity arbitrage check)
Suppose $S_0 = 40$, $K = 40$, $r = 5\%$, $T = 1$, $c = 4.00$, *and* $p = 2.00$. *Parity requires*

$$c + Ke^{-rT} = p + S_0.$$

The left side is

$$4 + 40e^{-0.05} = 42.05.$$

The right side is

$$2 + 40 = 42.00.$$

The apparent mispricing is 0.05 per share before transaction costs. In practice, this would likely be too small to exploit after bid-ask spreads and fees.

Example 10 (American put node decision)
At a node in an American put tree, $S = 38$, $K = 45$, $r = 4\%$, $\Delta t = 0.5$, $p = 0.55$, $f_u = 5$, *and* $f_d = 12$. *The continuation value is*

$$e^{-0.04(0.5)}[0.55(5) + 0.45(12)] = e^{-0.02}(8.15) \approx 7.99.$$

Immediate exercise gives

$$K - S = 45 - 38 = 7.$$

The node value is 7.99, so the put is not exercised at this node.

Example 11 (Delta in a tree)
At a node, the stock can move to 66 or 54. A call will be worth 9 after the up move and 1 after the down move. Its delta is

$$\Delta = \frac{9 - 1}{66 - 54} = \frac{8}{12} = 0.6667.$$

A trader short 100 calls would buy approximately 66.67 shares to create the one-step delta hedge.

Checklist for Students

After this lecture you should be able to:

- distinguish option rights from forward and futures obligations;
- define calls, puts, strike price, expiration, European exercise, and American exercise;
- compute payoffs and profits for long and short calls and puts;
- explain why option writers face margin requirements;
- distinguish option class, option series, intrinsic value, time value, and moneyness;
- describe how stock splits and stock dividends affect option contract terms;
- explain the roles of exchanges, market makers, brokers, clearing members, and the Options Clearing Corporation;
- list the main factors affecting stock option prices and sign their effects on calls and puts;
- derive and use upper and lower bounds for European and American options;
- derive put-call parity for European options on non-dividend-paying stocks;
- adapt parity and bounds for known dividends;
- explain why an American call on a non-dividend-paying stock should not be exercised early;
- explain why an American put may be exercised early;
- set up a one-step binomial tree and compute the risk-neutral probability;
- value a European option by risk-neutral expectation on a tree;
- construct the one-step delta hedge;
- value a two-step European option by backward induction;
- value an American option by comparing continuation and exercise values at each node;
- choose u, d, and p from volatility, the risk-free rate, and the time step;

- explain why the real-world expected stock return does not enter the no-arbitrage option price;
- explain why binomial prices converge as the number of steps increases.

Practice Questions

1. A call has strike 60 and premium 4. Compute the payoff and profit when the terminal stock price is 52, 60, and 71.
2. A put has strike 45 and premium 3.50. Compute the payoff and profit when the terminal stock price is 35, 45, and 52.
3. Draw the payoff diagrams for a long call, short call, long put, and short put. Label the strike and the break-even points after including premiums.
4. One equity option contract covers 100 shares. If a call is quoted at 2.75, what premium is paid for 12 contracts?
5. A stock is at 84. Classify a call with strike 80, a call with strike 90, a put with strike 80, and a put with strike 90 as in, at, or out of the money.
6. A put with strike 100 is priced at 13 when the stock is 91. Compute intrinsic value and time value.
7. A company has a 3-for-2 stock split. How should the terms of a call on 100 shares with strike 48 be adjusted?
8. Explain why a naked short call requires margin but a purchased call does not.
9. List the six main factors affecting stock option prices. For each, state whether an increase usually raises or lowers a call and a put.
10. A non-dividend-paying stock is 72, the strike is 70, $r = 4\%$, and $T = 0.5$. Compute the lower bound for the European call.
11. A non-dividend-paying stock is 38, the strike is 40, $r = 3\%$, and $T = 1$. Compute the lower bound for the European put.
12. A European call costs 6.20. The stock is 50, the strike is 52, $r = 5\%$, and $T = 0.75$. Use put-call parity to compute the corresponding European put price.
13. Suppose $c + Ke^{-rT} > p + S_0$. Describe the arbitrage trade and verify that the terminal payoff is nonnegative.

14. Why is it generally suboptimal to exercise an American call early when the stock pays no dividends?

15. Why can it be optimal to exercise a deep-in-the-money American put early?

16. A stock is 30 and can move to 36 or 24 in one year. A one-year call has strike 32 and $r = 5\%$. Compute p and the call price.

17. In the previous question, compute the hedge ratio Δ.

18. A two-step tree has $S_0 = 100$, $u = 1.1$, $d = 0.9$, $r = 4\%$, and $\Delta t = 0.5$. Compute the risk-neutral probability.

19. Use the tree in the previous question to value a European put with strike 100.

20. At an American put node, $S = 42$, $K = 50$, $r = 6\%$, $\Delta t = 0.25$, $p = 0.52$, $f_u = 5$, and $f_d = 11$. Should the option be exercised?

21. Compute u, d, and p for a stock option tree when $\sigma = 25\%$, $r = 4\%$, and $\Delta t = 1/12$.

22. Explain in words why risk-neutral probabilities are not generally the same as real-world probabilities.

23. If the number of tree steps is increased from 10 to 500 for a fixed option maturity, what happens to Δt? Why does this improve the approximation?

24. State clearly which parts of the source material are omitted from these notes and why.